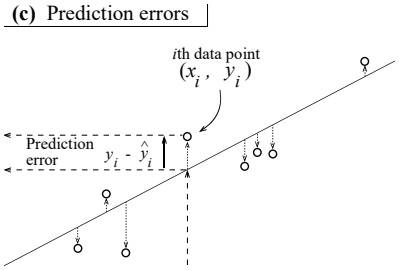
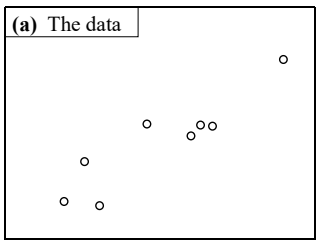
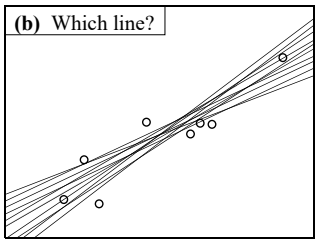
Fitting points to a line

- Goal is to get the "best" possible line
  - Estimated line: A + Bx
  - Minimize sum of squared vertical residuals





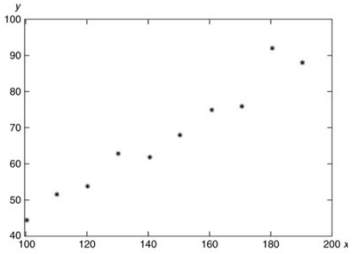


One (set of) outputs, one (set of) inputs

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Fitting points to a line

- Goal is to get the "best" possible line
  - Estimated regression line: A + Bx
  - Minimize sum of squared vertical residuals
- In this form, there is no mention of uncertainty
- Solution via linear algebra: minimize norm($Up - v$)
  - $U_{n\times 2}$: [**1** x_i]
  - $p_{2}x_{1}$ = $(A,B)^T$ (estimators)
  - $v_{n\times 1}$: Y_i

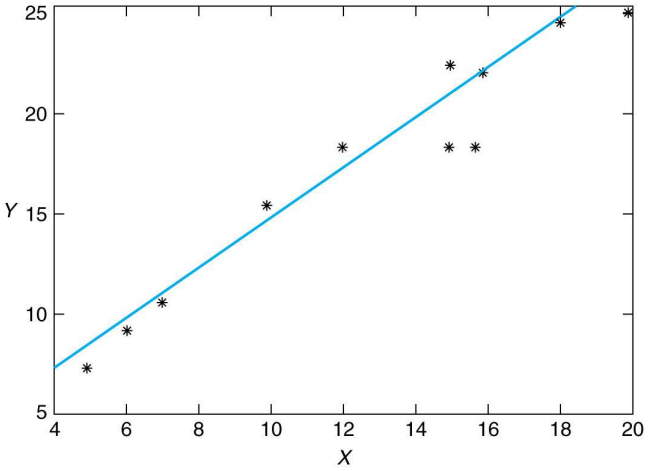


One output, one input

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Fitting points to a line



$$B = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\,(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

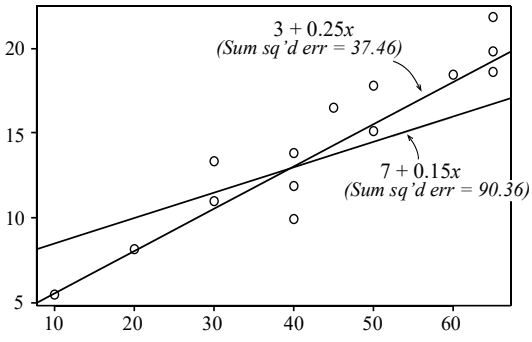
$$A = \bar{Y} - B\bar{x}$$

Proof on board

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Review



Prediction Errors

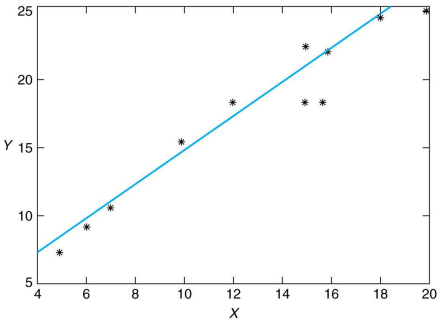
		3 + 0.25x		7 + 0.15x	
x	y	$\hat{y}$	$y-\hat{y}$	$\hat{y}$	$y-\hat{y}$
40	9.90	13.00	-3.10	13.00	-3.10
50	17.80	15.50	2.30	14.50	3.30
60	18.40	18.00	0.40	16.00	2.40
45	16.50	14.25	2.25	13.75	2.75
40	11.90	13.00	-1.10	13.00	-1.10
10	5.50	5.50	0.00	8.50	-3.00
30	11.00	10.50	0.50	11.50	-0.50
20	8.10	8.00	0.10	10.00	-1.90
50	15.10	15.50	-0.40	14.50	0.60
30	13.30	10.50	2.80	11.50	1.80
65	21.80	19.25	2.55	16.75	5.05
40	13.80	13.00	0.80	13.00	0.80
65	18.60	19.25	-0.65	16.75	1.85
65	19.80	19.25	0.55	16.75	3.05
	Sum of squared errors		37.46		90.36

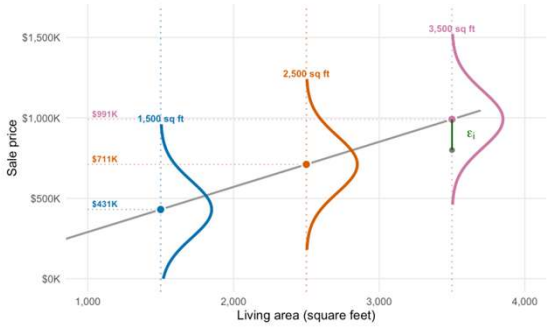
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Linear Regression

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$

β, α, x_i are assumed to be constant. We don't know β and α





$Y_i \sim \mathcal{N}(\beta x_i + \alpha, \sigma^2)$

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Least-squares fitting as MLE [1/2]

- Consider the linear model
  - $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
  - $p(y_i \mid x_i, \beta, \alpha) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(y_i - \beta x_i - \alpha)^2}{2\sigma^2}\right]$
  - $\ell(\beta, \alpha, \sigma) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta x_i - \alpha)^2 - n\log\left(\sigma\sqrt{2\pi}\right)$
- Maximizing likelihood gives the least-squares rule; applied to Y_i, it results in the normal equations

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \Pr(\mathbf{x}_{1\ldots I}|\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \left[\prod_{i=1}^{I} \Pr(\mathbf{x}_i|\boldsymbol{\theta})\right]$$

normal equations

$$\sum_{i=1}^{n} x_i Y_i = B\sum_{i=1}^{n} x_i^2 + A\sum_{i=1}^{n} x_i,$$

$$\sum_{i=1}^{n} Y_i = B\sum_{i=1}^{n} x_i + nA$$

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Least-squares fitting as MLE [2/2]

- Consider the linear model: $Y_i = \beta x_i + \alpha + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
- Solving the "normal" equations results in
- $\boxed{B = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}, \qquad \boxed{A = \bar{Y} - B\bar{x}},$
- where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i\,, \qquad \bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$

$$b_1 := \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{(\sum_{i=1}^{n} x_i y_i) - n\bar{x}\bar{y}}{(\sum_{i=1}^{n} x_i^2) - n\bar{x}^2} \left(= r\frac{s_Y}{s_X}\right) \qquad b_0 := \hat{\beta}_0 = \bar{y} - b_1\bar{x}$$

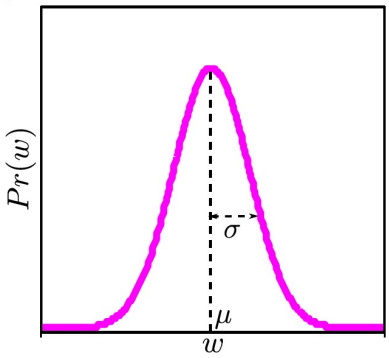
Formula: observed values of our intercept and slope estimators A,B.

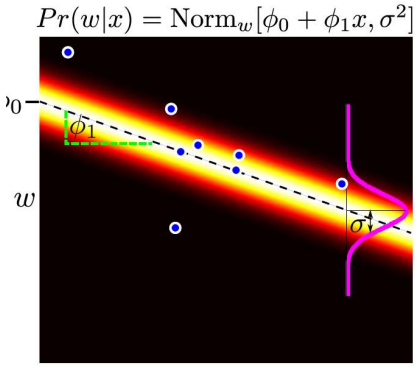
$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

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Linear Regression Review

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Most common case:
k=1

$$Pr(w_i|\mathbf{x}_i, \boldsymbol{\theta}) = \text{Norm}_{w_i}\left[\phi_0 + \boldsymbol{\phi}^T\mathbf{x}_i, \sigma^2\right]$$

Image notation: w is our response Y; ϕ_0=α, ϕ_1=β; ϕ is the coefficient vector.

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Regression Terminology

- Y is called output (response, dependent) variable
  - Also outcome, target, label, explained, predicted, regressand
- $(x_1, \dots, x_k)$ are called input or covariate or independent variables
  - Also predictor, feature, regressor, explanatory, attribute, factor, design, …
- We want to estimate the conditional density: $f(Y|\, x_1, \dots, x_k)$
  - $E = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_\mathrm{n}, y_\mathrm{n})\} = \{(x_{i_1}, x_{i_2}, \dots x_{i_k}, y_i) : i = 1, \dots, \mathrm{n}\}$

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